PROJECT TITLE : REVERSE PHARMA TRACKER

TEAM NAME : MERKLE TRACE

COLLEGE DOMAIN : LOGISTICS

PROBLEM STATEMENT

“Supply chains are brittle, opaque, and wasteful. Engineer the intelligence layer that makes goods move faster, cheaper, and greener”

Problems	Consequence
Expired drugs illegally re-enter the market	Counterfeit medicines reach patients; brand reputation destroyed; health emergencies
No verifiable proof of safe destruction	Regulators cannot trust paper certificates; illegal dumping of pharmaceuticals
Recalled batches cannot be reliably tracked	Recalled medicines remain in supply chain; patient safety at risk
Manual paperwork is prone to errors, loss, and manipulation	No single source of truth; audits take weeks; fraud goes undetected

Proposed Solution

“Blockchain-enabled end-to-end reverse logistics tracking with tamper-proof audit trails”

BLOCKCHAIN RECORD

1. Batch ID

Unique identifier for each batch

2. Expiry Date

Product validity tracking

3. Integrated Timestamp

Time of return initiation

4. Transfer Between Parties

Movement across stakeholders

5. Final Destruction Certificate

Proof of safe disposal

5-STEP BLOCKCHAIN FLOW

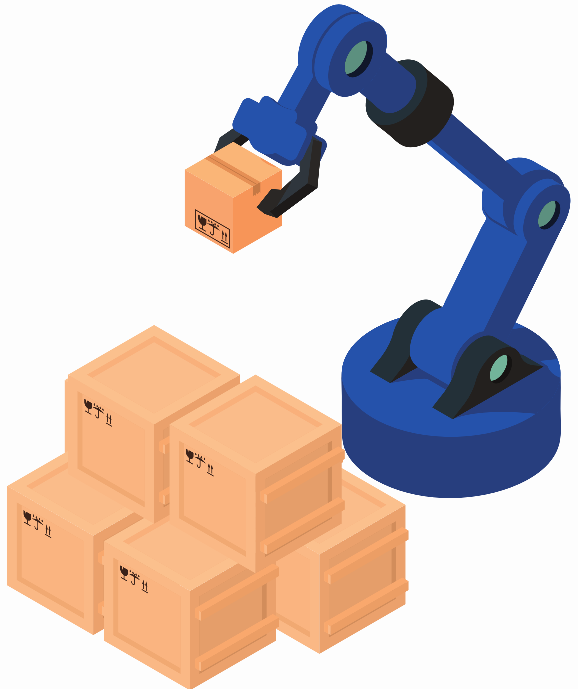
1. Retailer marks batch as expired

2. Entry is recorded on the blockchain

3. Distributor collects the batch → update logged

4. Manufacturer verifies → update logged

5. Destruction facility confirms → immutable record created



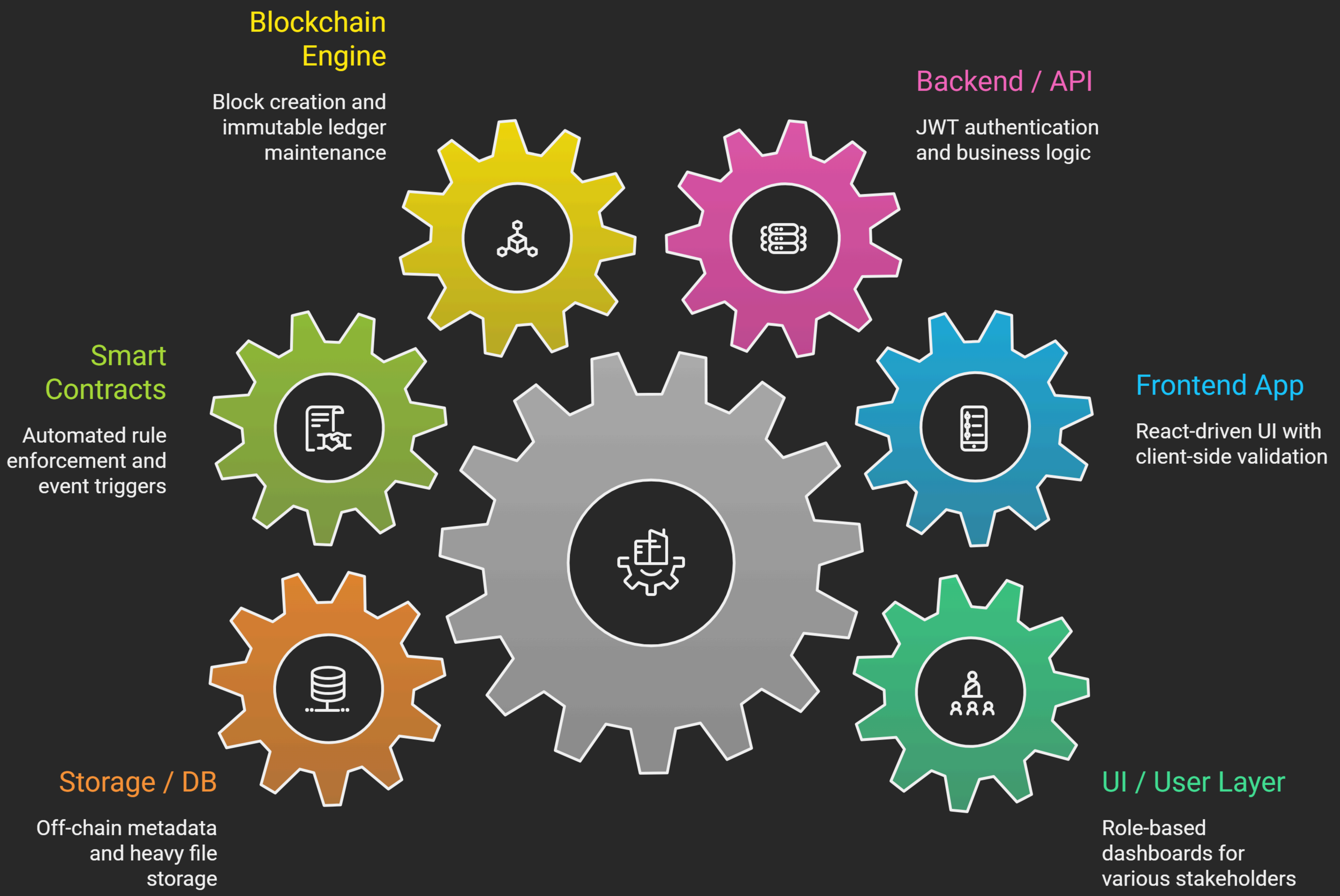
LITERATURE SURVEY

Title / Source	Methodology	Gap Identified
Evaluating Crowdfunding Campaigns (IEEE TEM, 2024)	Web Scraping, Text Mining	Focuses on funding, not physical goods tracking or destruction proof.
ProMark: Privacy-Aware Marketing (IEEE TDSC, 2025)	Blockchain, ZKPs	High computational cost; designed for marketing, not pharma disposal compliance.
Modeling Influencer Campaigns (IEEE TCSS, 2023)	Agent-Based Model	Simulates social spread, no integration with physical logistics or SCM.
Existing Pharma Logistics	Centralized Databases	Vulnerable to insider manipulation; no immutable audit trail for regulators.

Our Approach: A custom private blockchain + smart contract solution tailored for pharmaceutical reverse logistics.

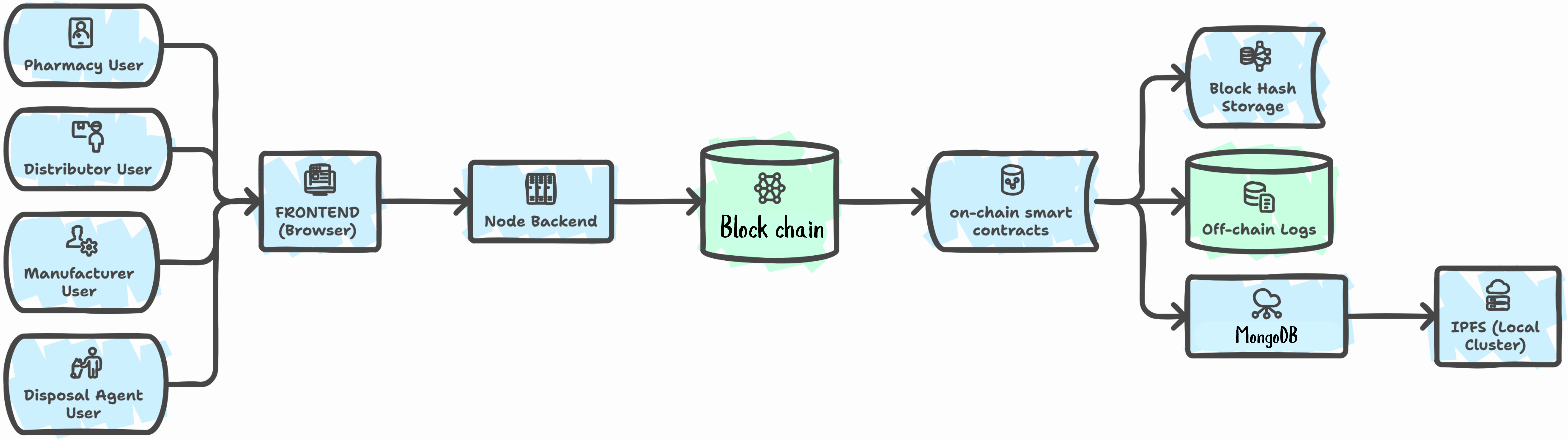
SYSTEM LAYERS

Pharmaceutical Supply Chain Layers

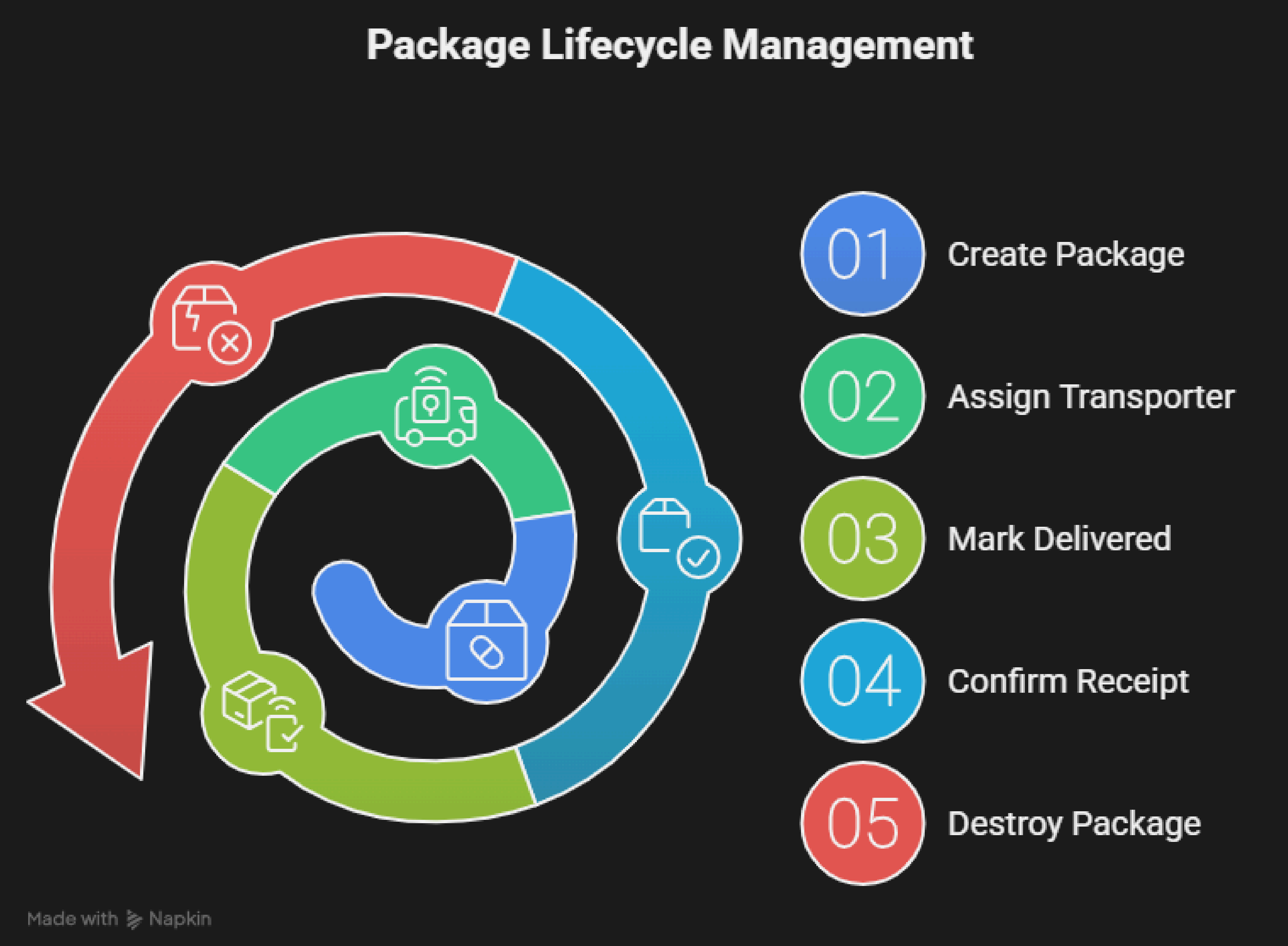


SYSTEM *Architecture*

Pharmaceutical Supply Chain and Disposal System



Architectural Flow Diagram (End-to-End)



Methodology

1. SHA-256 Hashing

- Used in: Block creation (Blockchain layer)
- Generates a unique hash for each block
- Ensures data integrity & detects tampering
- Links blocks securely

2. Merkle Tree Algorithm

- Used in: Inside each block (Data verification)
- Hashes transaction fields and combines them
- Produces a single Merkle Root

3. Blockchain-Based System Design

- Implemented a private blockchain network for secure data storage
- Blocks are linked using previous hashes to ensure immutability

4. Smart Contract-Driven Workflow

- Business logic is enforced using smart contracts (Solidity)
- Role permissions
- Valid state transitions
- Event logging

TECH STACK

FRONTEND

- HTML5 & CSS3
- JavaScript
- (Vanilla JS)
- qrcode.js
- jsPDF

BACKEND

- Node.js
- Express.js
- REST APIs

BLOCK-CHAIN

- Ethereum / solidity
- smart contracts
- web3.js

DATABASE

- MongoDB
- Block Hash Storage
- Off chain logs

CURRENCY
HOLDER

Network Name: Polygon Amoy (Testnet)

Chain ID: 80002
Currency Name: POL
Network: POLY AMOY

Thank You

Thank you for your time and attention.